Hussain Singapuri

AGENTIC AI & GENAI ENGINEER

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PROFESSIONAL SUMMARY

Agentic AI & GenAI Engineer with 3 years of professional software development experience at a US-based medical-device company, based in Pune. Builds **agentic AI** systems with **multi-agent orchestration** and tool calling (Python, asyncio, Claude API), **RAG** pipelines with embeddings, hybrid BM25 + vector search and **pgvector**, and **FastAPI** / REST services, with evaluation harnesses and guardrails. Built internal AI agents in production and an LLM inference engine from scratch (3.1x throughput over sequential decoding). Uses Claude Code daily. Claude Certified Developer and Orchestrator.

CORE COMPETENCIES

Agentic AI & Multi-agent Orchestration | RAG, Semantic Search & Vector Databases (pgvector) | Python, FastAPI & REST APIs | Tool / Function Calling | Prompt Engineering | LLM Evaluation & Guardrails | Docker & CI/CD (GitHub Actions) | Claude & Gemini APIs

WORK EXPERIENCE

NeuroNetworks AI Ed

2026 - Present

CTO & Principal Course Architect

Pune, India

- Lead technology for an Indian AI and tech education brand (Neuronetworks AI Ed Private Limited) offering programmes for students after Class 10
- Architected the curriculum: a 1-year AI Foundation course and five 3-year specialisations (AI/ML Engineering, Software Development, Data Science & AI, Product & UX / AI Design, Cybersecurity)
- Designing Neuromind Practical AI Lab, a browser-based IDE and sandbox with an AI tutor: phased spec, server sizing, engineering team plan and CAPEX/OPEX model
- Built and run the website neuronetwrks.com with a Python static-site generator deployed on Vercel, including programmatic SEO

US-based medical-device company

Mar 2023 - Sep 2026

Full Stack Developer

Remote

- Built **AI agents** for internal use
- Built and maintained full-stack web applications for the company
- Built several apps, including a CRM
- Performed penetration / security testing on 5+ of the company's mobile apps, all live on the Google Play Store

PROJECTS

multi-agent-research-crew

Open Source

Agentic AI: planner, parallel researcher, writer and verifier agents on a custom DAG orchestrator with retries, timeouts, token budgets and claim-level citation checking.

Python, asyncio, Claude

site-to-agent

Open Source

RAG: crawler, hybrid BM25 + vector semantic search on a pgvector vector database, fused with RRF, cited answers, SSRF-hardened fetching and an eval harness for answer quality.

Next.js, TypeScript, pgvector

nanoserve Open Source

Transformer LLM inference engine built from scratch: paged KV cache, prefix caching, continuous batching with preemption, speculative decoding, OpenAI-compatible streaming API. 3.1x throughput over sequential decoding in benchmarks.

Python, PyTorch, FastAPI

ai-lead-receptionist Open Source

LLM agent for home-service businesses: missed-call text-back, SMS and web chat that qualifies leads and books jobs, with an owner dashboard.

Next.js, TypeScript, Twilio, Claude, Supabase

lead-intel-engine Open Source

Discovers, enriches and scores local-business leads with a robots-aware, rate-limited crawler and transparent scoring.

Python, FastAPI, SQLite

contractor-ai-ops-dashboard Open Source

Multi-tenant analytics with signed event ingestion, funnel metrics, anomaly alerts and number-checked AI weekly summaries.

Next.js, TypeScript, Recharts, Postgres RLS

PageForge Portfolio project

Guardrails for LLM output: AI drafts only from approved facts and the account's public pages, a deterministic validator flags unsupported numbers, and a reviewer signs off claim by claim.

Next.js 16, React 19, Sanity, Supabase, Gemini

TripSync Project

AI business-travel planner: enter trip constraints, get three AI-generated packages, refine via chat, export to calendar.

Flutter, Supabase, Gemini structured outputs, GitHub Actions

More projects

Personal portfolio with full-stack backend, Orbital Surplus, Nightjar & Co., Lumen, Kilnwood Roasters, Cairnmoor, Lowlight Video Club, Safey Inventory

EDUCATION

Bachelor's, Computer Science (AI & Machine Learning) Vishwakarma Institute of Technology, Pune 2022-2025

CERTIFICATIONS

Claude Certified Developer	Anthropic, Foundations (CCDV-F)
Claude Certified Orchestrator	Anthropic, Foundations (CCAO-F)
Generative AI Leader	Google Cloud
Claude Code in Action	Anthropic
Claude Code 101	Anthropic

SKILLS

AI / LLM: AI agents, multi-agent orchestration, tool use / function calling, RAG (hybrid BM25 + vector, RRF), evals, prompt-injection defence, LLM inference serving, Claude API, Gemini API, Claude Code and AI coding agents

Languages: Python, TypeScript, JavaScript, SQL, Dart, HTML, CSS

Back end: FastAPI, asyncio, REST APIs, Node.js, webhooks (HMAC signing), Stripe Checkout and refunds, auth (scrypt, sessions, throttling), Resend

Data: pgvector, PostgreSQL, Neon, Supabase (RLS, Edge Functions), PGlite, SQLite, Sanity CMS

DevOps: Docker, GitHub Actions, Vercel

Security: mobile app penetration testing, CSP and security headers, SSRF hardening, input validation

Front end: Next.js, React 19, Tailwind CSS, shadcn/ui, TanStack Start / Query, GSAP, Lenis, three.js, WebGL, Canvas 2D, responsive and accessible UI, motion and interaction design

Automation & integrations: workflow automation, CRM integrations (HubSpot), PostHog, Twilio SMS, web crawling / scraping

Mobile: Flutter, Riverpod, Firebase **Design:** UI/UX design, design systems, technical SEO, JSON-LD structured data, programmatic SEO